

Athlete Nutrition Intelligence

Interactive Dashboard -- Design & Analysis Report

May 2026 | 15 Athletes | 5 Sports | Python + Chart.js

1. Project Overview

This project builds a fully interactive, browser-based nutrition intelligence dashboard for 15 elite athletes across five professional sports -- Tennis, Basketball, Formula 1, Soccer, and Breaking. The dashboard visualises body composition, macronutrient profiles, and training session load from a relational CSV dataset of 20 tables.

The interactive frontend (dashboard.html) is a single self-contained file built with Chart.js and vanilla JavaScript, requiring no server. The Python analysis layer (analysis.py) handles data loading, merging, and statistical summarisation using pandas, and produces this PDF report.

2. Dataset Description

The dataset is structured as a relational schema with 20 CSV tables:

- * Athletes: 15 elite athletes with name, DOB, sex, position
- * AthleteBodyMetrics: Weight, height, body-fat % per athlete
- * Sports / Teams / AthleteTeams: Sport taxonomy and team membership
- * Meals / MealItems: 30 meal records linked to food items
- * Foods / FoodNutrients / Nutrients: Nutrient composition per food serving
- * PerformanceSessions / PerformanceMetrics: Training sessions and output metrics
- * HydrationLogs / BeverageTypes: Fluid intake per session
- * FoodCategories / Seasons / DataSources: Reference / lookup tables

3. Key Findings

3.1 Body Composition by Sport

Sport	Avg Weight (kg)	Avg Body Fat %	Avg Height (cm)	Avg Session (min)
Basketball	101.0	10.0%	198	63
Breaking	65.3	8.0%	169	120
Formula 1	71.3	7.3%	178	90
Soccer	76.3	7.7%	178	110
Tennis	77.7	8.0%	186	120

3.2 Top Athletes by Protein Intake

Athlete	Sport	Protein (g/day)
Luka Doncic	Basketball	69
LeBron James	Basketball	67
Carlos Alcaraz	Tennis	46
Kylian Mbappe	Soccer	46
Max Verstappen	Formula 1	46

3.3 Hydration Totals per Athlete (ml / session)

Athlete	Total Volume (ml)
LeBron James	1000
Luka Doncic	1000
Cristiano Ronaldo	850
Kylian Mbappe	850
Jannik Sinner	850
Lionel Messi	850
Carlos Alcaraz	850
Stephen Curry	850

4. Dashboard Visualisation Design

Chart 1 -- Body Composition Intelligence (Bubble Chart)

Each athlete is plotted as a bubble: X-axis = body weight (kg), Y-axis = body fat percentage, bubble radius = height (cm). Colour encodes sport. Hovering any bubble triggers a custom HTML overlay card that reveals the athlete's name, sport badge, team, all three body metrics, daily calorie intake with a proportional fill bar, and macro gram breakdown -- giving deep insight into individual observations without cluttering the base chart.

Chart 2 -- Macronutrient Profiles (Radar Chart + Toggle)

A radar chart with four axes -- Calories, Protein, Carbs, Fat -- normalised to the maximum per axis so all sports fit on the same scale. Five sport-coloured toggle pills (Tennis, Basketball, Formula 1, Soccer, Breaking) animate their dataset in or out with Chart.js transitions. Active toggles glow in their sport colour; inactive ones dim to 40% opacity. Hovering any datapoint shows the actual gram/kcal value for that sport.

Chart 3 -- Session Analysis (Animated Bar Chart)

A vertical bar chart groups all 15 athletes by sport, separated by dashed dividers with emoji sport labels drawn via a custom Chart.js plugin. Bar opacity encodes intensity level (High = 0.82, Moderate = 0.55, Low = 0.32). A Duration / Calories tab pair animates a data swap, and the Replay button sequences bars in from zero one-by-one with a 75 ms stagger -- foregrounding the comparative load narrative.

5. Visual Design Choices

- * Pure-black base (#000000) with diagonal carbon-fibre grid (two overlapping 28 px diamond grids) and CRT scanlines for a street-racing aesthetic.
- * 90 animated speed-streak lines drawn on a background <canvas> element, using orange/red/white/blue gradients flying left-to-right at varied speeds and opacities.
- * Dual ambient neon blobs: orange fireball (bottom-left) and cold blue glow (top-right) that slowly breathe via CSS keyframes.
- * Angled clip-path: polygon(...) on cards and buttons gives a mechanical, armoured panel feel consistent across components.
- * Sport colours (Tennis #00D4FF, Basketball #FF6B2B, F1 #FF0044, Soccer #00E676, Breaking #D500F9) are used consistently across legend, radar, bubbles, and bar fills.
- * Space Grotesk (headings / UI) and Space Mono (numbers / monospace labels) from Google Fonts for a technical data-lab feel.
- * Hover on the bubble chart updates a background orb tint and slides in a rich tooltip card -- coupling micro-interaction with sport identity.

6. Technical Stack

Frontend:	HTML5, CSS3 (clip-path, backdrop-filter, keyframes), JavaScript ES2022
Charts:	Chart.js 4.x -- bubble, radar, bar chart types; two custom plugins
Fonts:	Google Fonts: Space Grotesk, Space Mono
Data layer:	Python 3, pandas, numpy
Report:	fpdf2 -- programmatic PDF generation
Hosting:	Single self-contained HTML file -- no server required

7. How to Run

Prerequisites: Python 3.9+, pip packages in requirements.txt

```
$ git clone https://github.com/junyanjune/nutrition-dashboard
$ cd nutrition-dashboard
$ pip install -r requirements.txt
$ python analysis.py          # generates writeup.pdf and prints summary
$ open dashboard.html        # or double-click -- works in any modern browser
```

No internet connection is needed to view the dashboard offline -- Chart.js and Google Fonts are loaded from CDN; save those locally for fully offline use.